

Register-recording for the diagonal W under full space-group symmetry

the general-symmetry analog of “compute $R_1 - R_2$ ”

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The analogy

Translation only. W is dense but block-Toeplitz: $W_{\mu\nu}(R_1 - R_2)$ depends on the *relative cell*. To index it you record one number:

$$|R_1, \mu\rangle |R_2, \nu\rangle |0\rangle \longrightarrow |R_1, \mu\rangle |R_2, \nu\rangle |R_1 - R_2\rangle.$$

Full space group. W is dense but constant on *space-group orbits of pairs*: $W_{g \cdot a, g \cdot b} = W_{a, b}$. The unique data ($\sum_i n_i^2$ values) is addressed by the **canonical orbit representative** of the pair — the “generalized $R_1 - R_2$.” Recording it is the same idea plus a fold by the $|P| = 48$ point ops.

Grid point (R, μ) sits at $R + r_\mu$. An op $g = \{A_g \mid \tau_g\}$ acts by $g \cdot (R, \mu) = (A_g R + \ell_\mu(g), \pi_g(\mu))$, where $A_g r_\mu + \tau_g = r_{\pi_g(\mu)} + \ell_\mu(g)$ defines the permutation $\pi_g = \text{perm}[g, \cdot]$ and fold-back vector $\ell_\mu(g)$. On the translation-reduced triple (μ, R, ν) (with R the relative cell) a point op acts as

$$g : (\mu, R, \nu) \longmapsto (\pi_g(\mu), \quad A_g \cdot R + \ell_\nu(g) - \ell_\mu(g) \pmod{\text{supercell}}, \quad \pi_g(\nu))$$

Registers

$ R_1\rangle \mu\rangle$	co-density 1: cell R_1 ($\lceil \log_2 N_k \rceil$), interp. point μ ($\lceil \log_2 n_{\text{IP}} \rceil$)
$ R_2\rangle \nu\rangle$	co-density 2
$ \cdot\rangle_\Delta$	relative-cell register, holds $R = R_1 - R_2$
$ \cdot\rangle_g$	coset-op register, $\lceil \log_2 P \rceil = 6$ qubits
$ \cdot\rangle_\kappa$	canonical-address register, holds the orbit rep $(\mu^\star, R^\star, \nu^\star)$
$ \cdot\rangle_\phi$	b -bit phase-gradient register (to apply W)

Step-by-step register state

0. Initial.

$$|R_1, \mu; R_2, \nu\rangle |0\rangle_\Delta |0\rangle_g |0\rangle_\kappa |0\rangle_\phi$$

1. Translation-reduce — the literal “ $R_1 - R_2$ ” step (a modular subtraction):

$$\longrightarrow |R_1, \mu; R_2, \nu\rangle |R\rangle_\Delta |0\rangle_g |0\rangle_\kappa |0\rangle_\phi, \quad R = R_1 - R_2.$$

The translation-invariant content is now the triple (μ, R, ν) .

2. Point-group canonicalize — the new part (a select/reduce over the 48 ops). For each $g \in P$ form the image triple $T_g = (\pi_g \mu, A_g R + \ell_\nu(g) - \ell_\mu(g), \pi_g \nu)$ (the boxed map; π, ℓ from the fixed

$|P| \times n_{\text{IP}}$ geometry QROM, A_g a signed coordinate permutation — nearly free), take the canonical (lexicographically minimal) T_g , write the achieving op into $|\rangle_g$ and the canonical triple into $|\rangle_\kappa$:

$$\longrightarrow |R_1, \mu; R_2, \nu\rangle |R\rangle_\Delta |g^\star\rangle_g |\mu^\star, R^\star, \nu^\star\rangle_\kappa |0\rangle_\phi.$$

Now $|\rangle_\kappa$ holds the **space-group orbit representative** — the unique-data address, the general-symmetry “ $R_1 - R_2$.”

3. Load and apply W . QROM addressed by $|\rangle_\kappa$ loads the one stored value $W(\kappa)$ (table size $\sum_i n_i^2$); add its angle into $|\rangle_\phi$ to apply the diagonal coefficient $e^{iW(\kappa)}$ (phase-gradient), then uncompute the load:

$$\longrightarrow e^{iW(\kappa)} |R_1, \mu; R_2, \nu\rangle |R\rangle_\Delta |g^\star\rangle_g |\kappa\rangle_\kappa |0\rangle_\phi.$$

4. Uncompute the recording. Reverse step 2 (same 48-op canonicalization) and step 1 (add R back), returning Δ, g, κ to $|0\rangle$:

$$\longrightarrow e^{iW(\kappa)} |R_1, \mu; R_2, \nu\rangle |0\rangle_\Delta |0\rangle_g |0\rangle_\kappa |0\rangle_\phi.$$

Net: the diagonal W applied on the pair, all work registers clean.

Why it works / costs

$W(\kappa)$ is one value per orbit, and *every* pair in the orbit maps to the same κ (pure permutation in real space $\Rightarrow$ no phase, $W(g \cdot a, g \cdot b) = W(a, b)$ exactly). So the reduced table has $\sum_i n_i^2$ entries and step 2 is exactly what routes each pair to its entry.

	translation only	full space group
recorded quantity	$R_1 - R_2$ (one lattice vector)	canonical orbit rep $(\mu^\star, R^\star, \nu^\star)$
compute cost	subtraction, $O(\log N_k)$	subtraction + 48-op canonicalize, $O(P \log N_k)$
extra tables	—	fixed $ P \times n_{\text{IP}}$ (perm, ℓ), n_k -indep.
W table size	$N_k n_{\text{IP}}^2$	$\sum_i n_i^2$ (unique data)

The only N_k -dependence is the $\log N_k$ register widths; $|P| = 48$ is a crystal constant. The 48-op canonicalize is the “register recording” term in the BSE resource estimate ($\approx$ stage-B reconstruction). Everything is reversible: $\kappa = \text{canon}(\mu, R, \nu)$ is a function of the inputs, so it is computed into ancilla and uncomputed after W is applied.

One subtlety

Nonabelian $\Rightarrow$ the orbit label is *not* a plain difference: the point ops also permute the endpoint indices $(\mu, \nu \rightarrow \pi_g \mu, \pi_g \nu)$ and fold cells $(\ell_\nu - \ell_\mu)$. So step 2 is a min over 48 images, not one subtraction — but the recorded variable is still just the relative configuration of the two points, now reduced to the point-group fundamental wedge instead of one cell.